Adobe Experience Manager



Highlights of Course Content	Course	Hours
	AEM Beginner	30
	AEM Advanced concepts	15
	AEM with Analytics and Target Integration	2
	AEM SPA & Forms & Assets	13
	AEM Authentication (OKTA , Azure AD with SAML)	2
	Groovy Script & GraphQL & ContextAware	3
	AEM Project Implementation	15
	Java	30
	Interview Questions - 150	
	Certification Practice Tests - ADO-E123,ADO-E134	
Total Hours (AEM (Live Training - 28, Recorded Sessions 52), Java 30)		110

Detailed TOC

Introduction to Adobe Experience Manager.

Adobe Experience Manager (AEM) is a content and digital asset management solution that empowers organizations to seamlessly create, organize, and deliver content across all digital touchpoints.

This comprehensive content management platform simplifies the often intricate process of handling various content assets, including documents, images, videos, PDFs, and online forms.

AEM redefines your online operations by offering a robust suite of features dedicated to rapid and efficient content creation, publication, and delivery. With Adobe AEM, you gain the agility to swiftly launch customer-facing websites, engage users through social pages that represent your brand, or establish content-rich blogs that cater to your audience's needs.

Table of content.

1. **AEM Introduction:**
   - Overview of Adobe Experience Manager (AEM)
   - History and evolution of AEM
   - Key features and benefits of AEM
2. **AEM Installation & WKND Project deployment:**
   - System requirements and prerequisites
   - Step-by-step installation process (Author & Publish)
   - Deploying the WKND Project (a sample project provided by Adobe)
3. **AEM Consoles:**
   - Introduction to AEM Consoles (CRXDE, Felix Console, etc.)
   - Purpose and functionalities of each console
4. **AEM Cloud Local Project Setup:**
   - Setting up a local development environment for AEM Cloud
   - Configuration and best practices for local development
   - Deploy newsportal project into author and publish servers.
   - Maven profiles and plugins for deployment.
5. **Components:**

- Understanding AEM components
- Creating custom components
- Component lifecycle and architecture
- Component properties
- Component Inheritance

6. **Dialog & Design Dialog:**
   - Introduction Granite and Coral UI
   - Creating dialog from scratch.
   - Component authoring and rendering.
7. **Editable Templates:**
   - Introduction to editable templates
   - Custom Template creation from scratch.
   - Difference between initial and structure modes.
   - What is template policy and how to configure it.
8. **Clientlibs:**
   - Clientlibs introduction
   - Clientlib properties
   - Managing client-side libraries in AEM
   - Creating and using clientlibs for CSS and JavaScript
   - Difference between embed and dependencies properties.
   - debugging clientlibs.
9. **Sling Request Processing:**
   - Understanding Sling request processing.
   - Request handling and resource resolution
   - Sling request selectors.
10. **HTL:**
    - Introduction to HTML Template Language (HTL)
    - Difference HTL and JSP.
    - HTL Expression and Operators
    - HTL block statements.
    - HTL Global Objects
    - HTL Context attribute
11. **Newsportal Project Setup (7h)**
    - Setting up a project

- Project structure and configuration
- Creating necessary directories for components, templates, client libraries, etc.
- Custom header and footer components.
- Recent Article component.
- Article Details Component
- Featured Articles Component
- In this category Component
- Tags Component

12. **Experience Fragment:**
    - Creating and managing experience fragments
    - Creating different variations in XF
    - Inheritance between variations
    - CF building blocks.
    - Reusing experience fragments across multiple pages
13. **Content Fragment:**
    - Introduction to content fragments
    - Creating Content Fragment models
    - Variations in CF
    - Export data in json format by using Asset HTTP API and GraphQL
14. **OSGi Introduction:**
    - Overview of OSGi (Open Service Gateway Initiative)
    - Bundle Introduction
    - Bundle Lifecycle
    - OSGI Architecture
    - Benefits and concepts of OSGi in AEM
15. **OSGi Components & Services:**
    - OSGI Annotations
    - OSGI Component lifecycle
    - Creating OSGi components and services
    - Injecting OSGI services into OSGI component/service.
16. **Runmodes & Configurations:**
    - Understanding runmodes in AEM
    - Different approaches to set custom runmodes.
    - Runmode specific configuration files.

- config files priority based on runmode matches.

17. **Sling Models:**
    - Introduction to Sling models
    - Sling Model annotations
    - Sling model injectors
    - Sling Model postconstruct annotation
    - Sling Model exporter.
18. **Sling Servlets:**
    - Sling servlet introduction
    - Different types of sling servlets.
    - Difference between SlingAllMethodServlet and SlingSafeMethodServlet.
    - Creating path based and resource based servlet.
    - Advantages of resource based servlet.
19. **Event Handlers:**
    - EventHandler Introduction.
    - Different type of events.
    - ResourceChangeListener
    - Get the resource resolver from system user.
    - ArticlePageActivationHandler implementation.
20. **API (WCM, Sling):**
    - Overview of AEM APIs
    - Sling API usage (ResourceResolver,Resource,ValueMap,ModifiableValueMap)
    - WCM API usage (PageManager,Page)
    - Perform CURD operations in users nodes by using sling API
21. **Schedulers:**
    - Creating and managing schedulers in AEM
    - Task scheduling and cron expressions
    - Scheduler with white board pattern and Scheduler API.
22. **Core Components:**
    - Introduction to AEM Core Components
    - Features of core components.
    - Extend core components
    - Style system in core components
23. **Workflows & User Management:**

- Creating users and groups.
- Add users into group.
- Add/Update permissions in group level.
- Introduction to AEM workflows
- Creating custom workflows and workflow models
- Workflow launchers usage

24. **Test Cases:**

- Junit 5 annotations.
- Sling mocks and Mockito
- Test cases for sling models
- Writing unit tests for AEM components and services
- Testing strategies and best practices

25. **Cloud Dispatcher:**

- Overview of Adobe Cloud Dispatcher
- Docker Installation.
- Apache web server set up and dispatcher deployment
- dispatcher.any configuration.
- vhost file configuration
- Rewrite rules and global variables
- Error page configuration.

26. **SQL 2:**

- SQL2 Operators
- AEM SQL2 queries
- Execute SQL2 with Java code.
- Query Optimization
- Sorting and Pagination
- Query Debugging

27. **Query Builder**

- XPath Introduction
- Query builder predicates.
- AEM Query builder examples.
- Execute predicates with java code.
- Query Optimization
- Sorting and Pagination

- Query Debugging

28. **Indexing**
    - Lucene & Solar Index Configuration
    - Custom Index Definitions
    - Index Maintenance
    - Reindex & Refresh
    - Performance Optimization
29. **Solar Search**
    - Lucene & Solar Index Configuration
    - Custom Index Definitions
    - Index Maintenance
    - Reindex & Refresh
    - Performance Optimization
30. **Okta & Azure AD Authentication**
    - OKTA & Azure AD Apps creation
    - Certificate Installation.
    - Truststore creation.
    - SAML authentication configuration.
    - Testing
    - Common SAML issues.
31. **AEM SPA**
    - AEM SPA project creation.
    - Deployment into Author and Publish.
    - Custom AEM component with React.
    - Sling Model with exporter.
    - AEM SPA Editor Configuration
    - Client-Side Routing
32. **AEM with Analytics + Target**
    - Adobe Launch Setup
    - Data Layer Implementation
    - Debugging launch events.
    - AEM with target Integration.
    - Different variations in target.
    - AB Testing.

- Mbox usage

33. GraphQL

- GraphQL setup in AEM
- Schema Definition
- Saved GraphQL quries.
- Search CF based on properties.
- Pass dynamic values to GraphQL.
- GraphQL predefined properies.

34. Groovy Script

- Groovy Script setup .
- Groovy Script operators and methods.
- AEM Console Scripting
- Content Manipulation
- Scheduled Tasks

35. Cloud Manager

- Environment Creation/Deletion/Modification.
- Prod/Non Pipeline Configuration
- Deployment Strategies
- Developer console for debugging.
- Admin console for user management.

36. Replication Agents

- Replication Agent Setup
- Reverse Replication agent
- Dispatcher flush.
- Replication common issue.
- Troubleshooting

37. Sling Content Distribution

- Forward Distribution
- Reverse Distribution
- Sync Distribution.

38. Sling RepoInit & Tagging

- Repository Initialization Scripts
- System user, users and groups creation.
- Content creation with RepoInit.

- Tagging Framework
- Tagging API.

39. Resource Merger

- Resource Merger properties.
- Override component dialog
- Use resource merger in page properties.

40. Richtext & Image Component

- Rich Text Editor Configuration
- Default RTE plugins.
- Different modes of RTE.
- Different types of Image dialog fields.
- file upload component.

41. Capstone Project

- Understanding project requirement.
- Dynamic components.
- Youtube integration.
- Fetch dynamic data with SQL2.
- Final Presentation

42. Newsportal

- Project setup.
- Dynamic navigation and footer component.
- Article page dynamic components.

43. Context Aware Config

- Introduction.
- Context Aware config java file creation.
- context aware configuration nodes.
- link config with content.
- read Context aware config in sling models.
- Use Cases

44. AEM Assets

- Asset Introduction
- AEM Cloud assets processing.
- Metadata Schemas
- Custom Metadata Profiles.

- Custom Search filter.
- HTTP Asset API

45. **AEM Forms**
    - Forms Introduction
    - Forms project setup and deployment.
    - OOTB forms components.
    - DB connection.
    - JSON and XSD schema forms.
    - Form data models.
    - Guidebridge API
46. **Java**
    - OOPs
    - Fundamentals
    - Exception Handling
    - Collections



AEM Overview

Adobe Experience Manager (AEM) is a comprehensive content management solution for building websites, mobile apps, and forms, aimed at helping businesses manage their marketing content and digital assets effectively. Here's an overview of AEM's key features and capabilities:

Content Management System (CMS)

- AEM provides a robust platform for creating, managing, and optimizing digital experiences across all customer touchpoints. With its user-friendly interface, marketers and content creators can easily manipulate content without needing deep technical expertise.

Digital Asset Management (DAM)

- Integrated with AEM, the DAM system organizes, stores, and retrieves media files efficiently while ensuring version control and rights management. This is particularly useful for enterprises that handle large volumes of digital media.

Multi-Site Management

- Enables organizations to manage multiple sites across different regions and languages from a single interface, simplifying global operations and ensuring brand consistency.

Personalization

- Utilizes Adobe Target to deliver personalized experiences to users based on their behavior, device use, and past interactions with the brand. This capability helps in increasing engagement and conversion rates.

Cloud Services

- Adobe offers AEM as a Cloud Service, providing scalability, security, and flexibility without the overhead of managing the infrastructure. This model ensures that the platform is always up to date with the latest features and security measures.

Integration Capabilities

- AEM seamlessly integrates with other Adobe solutions like Adobe Analytics, Adobe Campaign, and Adobe Creative Cloud, offering a unified platform to handle marketing, analytics, and creative workflows effectively.

Workflow Engine

- The platform includes a workflow engine that automates the process of reviewing and approving content, thus speeding up content delivery and maintaining consistency.



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